

Reverse-Engineering Epistemology

A Systematized Framework for Multi-Paradigm Knowledge Acquisition

Deconstructing modern information vectors requires moving beyond linear absorption into algorithmic processing. Based on the structural analysis documented in *image.png*, learning strategies must adapt dynamically to the intrinsic axiomatic density of the subject matter. This framework formalizes the operational divergence between loose topological networks (Liberal Arts) and strict dependency chains (Hard Sciences), synthesizing them into a unified execution paradigm for deep knowledge domain acquisition.

1. The Strategic Taxonomy of Knowledge Domains

Knowledge structure is governed by its degree of axiomatization. General **Liberal Arts** exhibit a flat layout with weak sequential dependency, permitting random-access, non-linear navigation directly to high-interest nodes. However, as domains scale in complexity into **Deep Humanities** (e.g., Analytic Philosophy, Mathematical Linguistics), they shift into pseudo-axiomatic systems where key definitions act as strict prerequisites.

Conversely, **Hard Sciences** are structured as rigid, high-density linear dependency graphs. Every subsequent theorem T_n functions as a pure logical output derived from preceding axioms $A_0, A_1, \dots$ and operational lemmas. Skipping developmental dependencies triggers immediate cognitive runtime compilation errors.

2. Paradigm-Specific Cornell Note Adaptations

Standard linear note-taking generates an illusion of competence via passive transcription. True compilation requires a spatial layout optimized for active recall and structural mapping. As illustrated in *image.png*, the Cornell note architecture adapts fundamentally across paradigms:

- **The Core Invariant:** Never copy verbatim text from reference materials. Verbatim copying shifts cognitive loads to muscle memory, paralyzing active processing.
- **Liberal Arts Mode (Topological Mapping):** The left-hand column acts as an abstract cue tracker for core concepts, while the right-hand column records high-density, highly compressed definitions formulated purely in personal logical terms.
- **Science Mode (Algorithmic Execution):** The left-hand column lists absolute theorems and definitions. The right-hand column rejects pure passive summaries, serving instead as a ledger for **Tactics** (meta-strategies, cognitive friction points, and structural invariants). The execution phase demands an external draft pad for manual mathematical/algorithmic derivation.

3. Cross-Paradigm Methodological Matrix

Dimension	General Liberal Arts	Hard Sciences	Deep Combined Domains
Structural Setup (Catalog Check)	Isolate interest anchors; establish broad conceptual topology.	Map the dependency tree; verify absolute prerequisite coverage.	Extract the core micro-axioms and trace their global macro- resonance.
Reading Architecture	Non-linear, random-access routing across decoupled sections.	Strict linear progression from base principles (A_0 $\rightarrow T_n$).	Iterative loops: switch between deductive rigor and network validation.
Information Compression	Strip rhetoric/examples; capture the invariant conceptual core.	Isolate operators; strip symbolic notation to expose raw logic.	Pierce academic jargon; deconstruct formulas into underlying sandboxes.
Cornell Space Allocation	Left: Abstract Core Concepts. Right: Compact Self- Language Explanations.	Left: Rigorous Theorems / Constraints. Right: Tactical Execution Blueprints.	Macro-level Cornell mapping for systems architecture paired with rigorous draft derivations.
External Hardware Extension	Minimal. Internal working memory manages the network graph.	Mandatory draft pads / algorithmic simulation sandboxes.	Dual-track: dynamic rough scratchpads validated via structured ledgers.
Internalization Standard	Unassisted synthesis and dynamic extension of core arguments.	Flawless derivation of target theorems from a closed book state.	Coherent sandbox execution; seamless transition between math and concept.

System Criterion for Deep Knowledge: At frontier layers, the artificial split between qualitative networks and quantitative computations collapses (*Deep Knowledge* $\equiv$ *Combined Field*). Advanced domains demand a dual-engine cognitive architecture: applying rigorous mathematical computation to trace the exact trajectories of highly abstract, multivariable conceptual sandboxes.